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Paleomagnetic Constraints on the Origin and Drift History of the North Qiangtang Terrane in the Late Paleozoic

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Key Points:
- The North Qiangtang terrane (NQT) was located at equatorial paleolatitude (−7.6 ± 5.6°N) during the Late Permian (~259 Ma)
- The NQT drifted northward together with the South China block (SCB) during the Permian
- The NQT rifted from the northern margin of Gondwana during the Devonian

Supporting Information:
- Supporting Information S1
- Table S1
- Table S2
- Table S3
- Table S4

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Abstract To better constrain the origin and drift history of the North Qiangtang terrane (NQT), we report a well-dated paleomagnetic pole from the Late Permian volcanics of the NQT that appears to average out secular variation. Our new results yield a paleolatitude of −7.6 ± 5.6°N at ~259 Ma for our sampling area, which confirms the NQT drifted northward during the Permian and Triassic periods. The equatorial paleolatitude of the NQT is similar to that of the coeval South China block, demonstrating that they were in close proximity. Combined with palaeontological and magmatic evidence, paleomagnetic constraints on the drift of the NQT in the Permian indicate that the NQT moved northward together with the South China block at this time. The paleolatitude evolution of the NQT implies that the NQT rifted from the northern margin of the Gondwana in the Devonian, which is earlier than the departure time of the South Qiangtang terrane.

Plain Language Summary The Tibetan Plateau is composed of several different blocks that accreted to the southern margin of Asia. There are still several unanswered questions, such as the following: Where did these blocks originate? How did these blocks accrete to Asia? How did the oceanic basin evolve? In this work, we provide robust evidence to show that the Northern Qiangtang was located at equatorial latitude (−7.6 ± 5.6°N) during the Late Permian (~259 Ma). The northward drift history together with features of the Northern Qiangtang and South China block indicates that they moved northward together during the Permian and that the Northern Qiangtang rifted from the northern margin of the Greater India margin of Gondwana during the Devonian.

1. Introduction

The Phanerozoic evolution of the Tethyan realm involved successive dispersion, subsequent northward drift, and eventual Asian accretion of a series of Gondwana-derived terranes (Metcalfe, 2013). The northward movement of these Gondwana-derived terranes, which was accompanied by the opening or closing of the Paleo-Tethys, Meso-Tethys, and Ceno-Tethys, determined the distribution of the land and sea and affected the regional and even global climate dynamics (Metcalfe, 2013). Quantifying the rift and drift history of these terranes is critical for reconstructing the distribution of land and sea in the Tethyan realm and understanding the plate tectonics history of these terranes (Song et al., 2017). For example, the Qiangtang terrane is separated from the Songpan-Ganzi terrane to the north by the Triassic Jinsha suture zone (JSS), which marks the closure of Paleo-Tethys, and from the Lhasa terrane to the south by the Jurassic-Cretaceous Bangong-Nujiang suture zone by the closing of Meso-Tethys (Figures 1a and 1b; Yin & Harrison, 2000). Therefore, the rift and drift histories of the Qiangtang terrane play a critical role in understanding the evolution of the Paleo-Tethyan and Meso-Tethyan Oceans.

Currently, it is widely accepted that the Qiangtang terrane can be further divided into the North Qiangtang terrane (NQT) and the South Qiangtang terrane (SQT) by the Qiangtang Metamorphic Belt or the Longmuco-Shuanghu suture zone due to the discovery of the eclogite, blueschist, and ophiolitic mélange (Li et al., 2006). The Qiangtang metamorphic belt trends from west to east in the center of the Qiangtang

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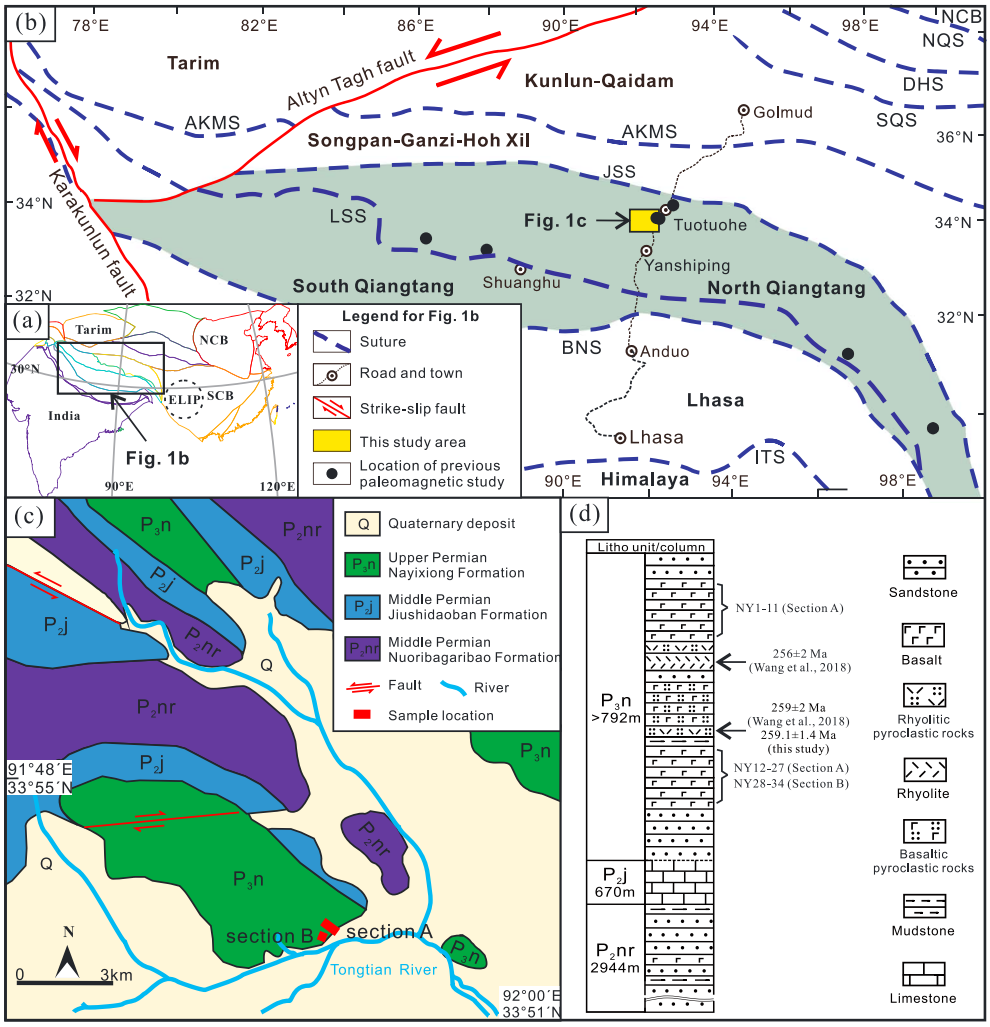


Figure 1. (a and b) Regional geological map of the Qiangtang terrane and adjacent areas (modified from Yin & Harrison, 2000). The abbreviations of the tectonic units are AKMS, Ayimaqin-Kunlun-Muztagh suture zone; BNS, Bangong–Nujiang suture zone; DHS, Danghe Nan Shan suture zone; ELIP, Emeishan large igneous province; ITS, Indus-Tsangpo suture zone; JSS, Jinsha suture zone; LSS, Longmuco-Shuanghu suture zone; NCB, North China block; NQS, north Qilian suture zone, and SQS, south Qilian suture zone. (c) Geological map of the sampling area (modified from Li, 2015). (d) Stratigraphic column of the Permian strata showing our sampling unit (modified from Wang et al. (2018) based on our observations).

terrane (Kapp et al., 2000). This metamorphic belt was sometimes considered to have been underthrust from the JSS and exhumed in the interior of a single *Qiangtang Terrane* (Kapp et al., 2000). However, most researchers suggest that the Qiangtang terrane was formed by the suturing of two independent terranes along the Longmuco-Shuanghu suture, which extends from northwest to southeast across the Qiangtang terrane (e.g., Zhang et al., 2014, 2016). Division of the NQT and SQT can be further supported by independent evidence from paleontological studies. The Late Paleozoic strata of the NQT contain abundant warm-water fossils, indicating a Cathaysian affinity (Zhang et al., 2013), whereas the Permo-Carboniferous strata of the SQT are characterized by glaciomarine deposits and cold-water biota, implying a typical Gondwana affinity (e.g., Zhang et al., 2013).

A wealth of geological evidence including paleobiogeography records has been used to constrain the origin and past positions of the NQT. However, they yielded contradictory interpretations (Song et al., 2017; Zhang et al., 2013). For example, Huang et al. (1992) argued that the NQT had drifted away from the SCB by the Late Permian because of its Cathaysian affinities in paleobiogeography during the Late Paleozoic. Metcalfe (2002, 2013) advocated that the NQT rifted from the SCB by back-arc spreading in the

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Carboniferous. Recently, Song et al. (2017) suggested that the NQT rifted away from Gondwana prior to the Permian.

The drift history during the Late Paleozoic is significant to understanding the origin of the NQT. To better constrain the drift history of the NQT and its relationship with adjacent terranes, as well as the evolution of the Tethys Ocean, we present a well-dated, high-quality paleomagnetic pole from the Late Permian Nayixiong Formation (Fm) volcanics of the NQT and deduce its further tectonic implications.

2. Geological Setting and Sampling

2.1. Geological Setting

The strata of the NQT comprise Late Devonian-Triassic and Jurassic-Cenozoic sedimentary rocks, as well as Paleozoic to Mesozoic volcanic rocks (e.g., Li, 2015). The Permian volcanic and sedimentary rocks are widely distributed in the eastern half of the NQT. These Permian volcanics have been attributed to the northward subduction of the Paleo-Tethys Ocean. Some of the volcanics may also be influenced by the Emeishan mantle plume (Liu et al., 2016).

Our study area, that is, the Zhouqiongmalu area, which is ~40 km north of the town of Yanshiping (Figures 1b and 1c), is located on the south of the JSS. The Permian strata can be divided into three formations in this area (Figure 1d). They are, from bottom to top, the Nuoribagaribao Fm (P_2nr), Jiushidaoban Fm (P_2j), and Nayixiong Fm (P_3n). The Nayixiong Fm consists of basalts, basaltic pyroclastic rocks with subordinate rhyolites and rhyolitic tuffs, and fine-grained sandstones intercalated with minor tuffs. The *Palaeofusulina parafusiformis-Nanlingella simplex* fusulinid assemblage recognized in the Nayixiong Fm indicates a Wuchiapingian (Late Permian) age (He et al., 2009). The regional structural relationships between the Permian and Triassic strata suggest that the fold of the Late Permian Nayixiong Fm occurred during the time interval from the Late Permian to the Late Triassic (Li, 2015), mainly due to the Triassic collision between the NQT and the SQT (Dan, Wang, White, et al., 2018).

2.2. Sampling

In total, 34 paleomagnetic sites were collected from two sections in the sampling area (Figure S1a in the supporting information). These two sections were chosen at two limbs of the anticline for a possible fold test. Twenty-seven sites and seven sites of volcanic rocks were collected from sections A and B, respectively (Figure S1). Seven sites from section A (33.9°N, 91.9°E) were collected as standard paleomagnetic cores using a portable gasoline-powered drill, and these cores were oriented by both a magnetic compass and a Sun compass. The declination difference is less than 2° between these two orienting methods, suggesting that the local magnetic disturbance can be neglected. Because the portable drills were broken, the remaining 27 sites were collected as oriented blocks. Each sampling site spans several meters of stratigraphic thickness or at least one lava flow. The bedding attitude of both sampling sections is clear (Figure S1). Additionally, one block sample of rhyolite was collected from the middle part of section A for zircon U-Pb chronology (Figure 1d). See Text S1 in the supporting information for laboratory techniques and measurements (Andersen, 2002; Cogné, 2003; Kirschvink, 1980; Lowrie, 1990; Ludwig, 2003; Xie et al., 2008; Xiu et al., 2001).

3. Results

3.1. Zircon U-Pb Geochronology

Most of the zircons from the rhyolite are euhedral to subhedral prisms with a width no smaller than 40 μm (Figure S2). The Th/U ratio is larger than 0.5, indicating that these zircons have a magmatic origin (Table S1 in the supporting information). Sixteen of the 21 zircons have concordant age (Table S1). They reliably show similar $^{206}Pb/^{238}U$ ages (Figure 2a), yielding a weighted average age of 259.1 ± 1.4 Ma with mean square weighted deviation = 0.77 (Figure 2b). The age of ~259 Ma determined by laser ablation sampling and multiple-collector inductively coupled plasma mass spectrometry (LA-MC-ICPMS) dating method is well consistent with the secondary ion mass spectrometer (SIMS) U-Pb zircon dating results (~259 Ma) for the same volcanic rock from the same bedding (Wang et al., 2018). Therefore, we confidently conclude that the Nayixiong Fm volcanics most likely erupted at ~259 Ma.

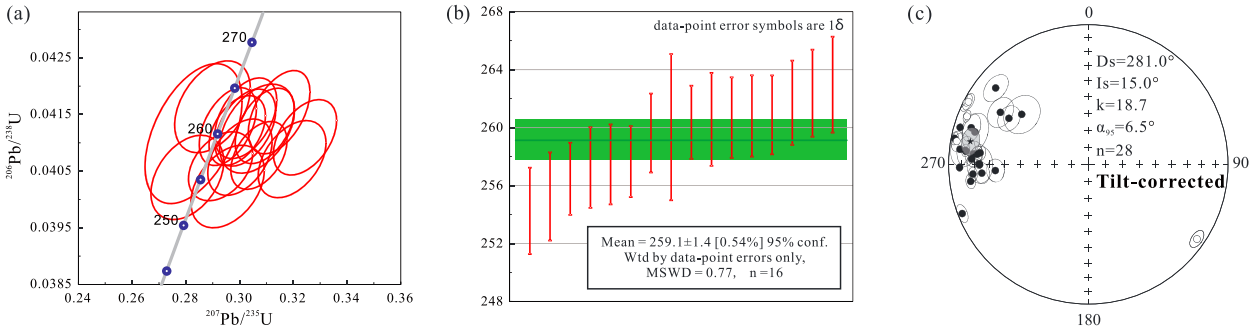


Figure 2. (a) Concordia diagrams showing the $^{206}Pb/^{238}U$ ratios in the zircon grains. (b) Weighted average of the apparent $^{206}Pb/^{238}U$ ages. (c) Equal-area projections of the site-mean directions of the Nayixiong Fm volcanic rocks. The star indicates the overall mean direction.

3.2. Paleomagnetic Results

Rock magnetism and magnetic petrography are described in Text S2 (Roberts et al., 1995). Stepwise thermal demagnetization was carried out to clean the magnetic remanence. The typical zijderveld plots of Nayixiong Fm volcanic rocks from two sections are shown in Figure S5 (Zijderveld, 1964). Some specimens have no clear decay of the remanence intensity until a stepwise demagnetization up to ~650 °C, indicating that the main magnetic carrier is hematite (Figure S5). Other specimens have a low-temperature component below ~450 °C, which is close to the present-day geomagnetic field direction in geographic coordinates, implying a recent viscous remanent magnetization. For most specimens, a high-temperature component (HTC) decaying toward the origin can be isolated between ~450 °C and ~580/680 °C. The same remanence directions carried by magnetite and hematite indicate that they were formed by exsolution during initial cooling (Geissman & Van der Voo, 1980; Song et al., 2015; Figure S5).

The HTC directions were used to calculate the site-mean directions (Text S3 and Tables S2 and S3). The final mean direction of the 28 sites is Dg = 276.5°, Ig = 10.2°, kg = 16.1, and α_{95} = 7.0° in situ and Ds = 281.0°, Is = 15°, ks = 18.7, and α_{95} = 6.5° after tilt correction (Figure 2c and Table S3). These 28 site-mean directions pass the McFadden (1990) fold test at 95% and 99% confidence levels (i.e., ξ_1 = 9.62 before, and 1.4 after, tilt correction; ξ_2 = 12.98 before, and 3.18 after, tilt correction; and critical value ξ = 6.16 at the 95% confidence level and 8.70 at the 99% confidence level), indicating that the characteristic remanent magnetization directions record a pretilting magnetization. The reversal test for the site-mean directions is indeterminate, which may be due to only one site showing different polarities. Nevertheless, its site-mean direction is antipodal to that of some sites (Figures 2c and S5j and S5k). Our new results fulfill all seven quality criteria proposed by Van der Voo (1990) and yield a Fisher mean virtual geomagnetic pole (VGP) at 13.6°N, 2.4°E with A_{95} = 5.6°. See Text S4 for details (Yan et al., 2016).

4. Discussion and Conclusions

4.1. Paleosecular Variation

Secular variation must be averaged out when volcanic-based paleomagnetic data can be used for tectonic reconstructions. Because the stratigraphically adjacent volcanic sites with the same remanence directions may record the same *spot reading* of the paleomagnetic field (e.g., Lippert et al., 2014), our paleomagnetic data set is evaluated for directional independence to average out secular variation by using a common direction test (McFadden & McElhinny, 1990). Only two adjacent volcanic sites share similar directions within the 28 sites, indicating that the difference between the mean direction of the 28 sites and the true mean direction can be neglected. The angular dispersion (S = 16.4°) of the corresponding VGPs of the 28 sites is similar to the dispersion of S_λ = 17.4° (limits between 12.5° and 22.5°) observed in VGP populations from 0 to 5 Ma lavas at a similar latitude (Johnson et al., 2008; Table S3). The A_{95} (5.6°) of the mean VGP is well within the N-dependent error envelope (3.2°, 10.0°) proposed by Deenen et al. (2011). These results suggest that our new data set has averaged out paleosecular variation and thus can be used for tectonic reconstruction.

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4.2. Position of the NQT During the Late Permian

Previous paleomagnetic investigations of Late Permian rocks of the NQT have been performed in the Raggyorcaka area by Cheng et al. (2012), in the Tuotuohe area by Cheng et al. (2013), and in the Tuoba area by Huang et al. (1992). Cheng et al. (2012) and Cheng et al. (2013) reported two paleopoles from five sedimentary sites and five volcanic sites, respectively, and concluded that the NQT was located at different paleolatitudes of approximately 13.1 and −7.7°N in the Late Permian (Table S4). Other Late Permian paleomagnetic results of 10 sedimentary sites reported by Huang et al. (1992) positioned the NQT at ~4.8°N (Table S4). All the paleomagnetic results contain no more than 10 site directions in each study. The secular variation of paleomagnetic field cannot be fully averaged out by only five volcanic sites (Cheng et al., 2013). Besides, the possible inclination shallowing have not been tested and corrected for sedimentary data by Cheng et al. (2012) and Huang et al. (1992). Moreover, no isotopic ages are available to constrain the precise ages of these poles. Nevertheless, these pioneering investigations constrain the NQT located at or near the equator in the tropics during the Late Permian.

The Nayixiong Fm paleomagnetic data of our new results are from 28 volcanic sites without inclination shallowing and fulfill the quality criteria of Van der Voo (1990), Deenen et al. (2011), and Johnson et al. (2008). In addition, previous studies on the Nayixiong Fm volcanic rocks obtained the Fisher (1953) mean direction of five sites at Ds = 250.1°, Is = 14.9°, ks = 37.3, and α_{95} = 12.7° after tilt correction (Cheng et al., 2013). The inclination (corresponding to paleolatitude) is nearly equal to that calculated from the 28 sites in our new data. Given that the site-mean directions reported by Cheng et al. (2013) include directions determined using the remagnetization great circle plane fitting techniques, which are not as reliable as the HTC directions determined using principal component analysis, we only used our new data for tectonic reconstruction.

Notably, our paleomagnetic results of the Nayixiong Fm volcanics revealed that 149 (27) of 184 tilt-corrected directions have a downward (upward) inclination with a westward declination. When a (sub) equatorial paleolatitude is observed, the choice between normal and reversed polarity is often ambiguous (Huang et al., 1992). The age of the middle part of our sampling section is dated at ~259 Ma, which is consistent with the age (~259–256 Ma) determined by SIMS U-Pb zircon dating (Wang et al., 2018) and has reversed polarity (Ogg et al., 2016). Therefore, we concluded that our sampling area was more likely located at a paleolatitude of −7.6 ± 5.6°N at ~259 Ma. Given the accuracy and uncertainty of analytical techniques in zircon U-Pb geochronology, we cannot fully exclude the possibility of normal polarity characteristic remanent magnetization directions for most of the volcanic sampling sites (27 sites) in this study. In this case, the NQT would be located at 7.6 ± 5.6°N in the Late Permian. Nevertheless, in either case, our new paleomagnetic results reveal a (sub) equatorial paleolatitude in the Late Permian for the NQT.

4.3. The Northward Movement of the NQT

Few paleomagnetic results from the Permian and Triassic rocks from the Qiangtang terrane have been reported, mainly due to a lack of proper sections for paleomagnetic sampling (Cheng et al., 2012, 2013; Lin & Watts, 1988; Song et al., 2012; Yang et al., 2017; Ye & Li, 1987). We have summarized available data in Table S4 in supporting information (Figure 1b). To obtain reliable reconstruction, we only used these data, which fulfill the following criteria, as Song et al. (2017) have applied: (1) positive field tests that constrain the age of magnetization, (2) the observed paleolatitude is calculated from 10 or more site/group mean directions for volcanic rocks or from 100 or more sample directions for sedimentary rocks, and (3) the A_{95} for the study passes the Deenen et al. (2011) criteria. Only three paleomagnetic results from Song et al. (2015, 2017) and this study fulfill these criteria (Figure 3 and Table S4). Song et al. (2015, 2017) obtained high-quality, well-dated paleomagnetic results of the Early Permian and Late Triassic ages, which come from many volcanic sites with clear bedding attitudes. These high-quality paleomagnetic results have shown that the NQT was located at −22.1 ± 4.7°N at ~297 Ma and 31.5 ± 3.0°N at ~209 Ma for the reference point at (33.9°N, 91.9°E). This paleolatitude evolution reveals a ~53.6° latitudinal (~5,900 km) northward movement in the Tethys Ocean from the tropical paleolatitude of the southern hemisphere in the Early Permian to the subtropical paleolatitude of the northern hemisphere, until it collided with the North China block and Tarim in the north (Song et al., 2015, 2017).

The observed paleolatitude of the Late Permian further supported this gradual northward movement and showed that the NQT was located at −7.6 ± 5.6°N at ~259 Ma (Figures 3a and 3d). Notably, the

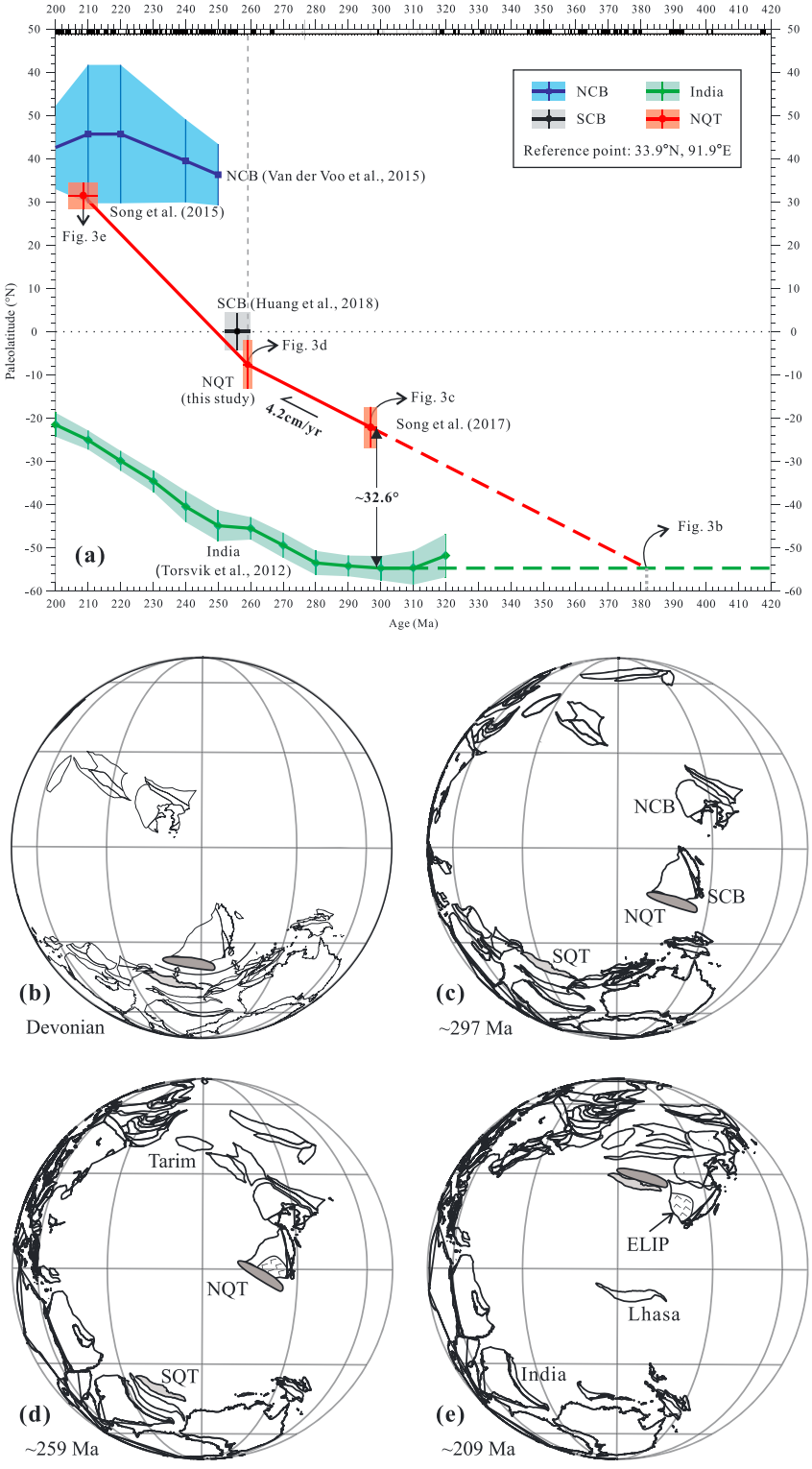


Figure 3. (a) Observed and expected paleolatitude evolution of the North Qiangtang terrane (NQT) and adjacent blocks (including North China block, South China block, and India) during the Devonian-Triassic period. The paleolatitudes were calculated for the reference point at (33.9°N, 91.9°E). The expected paleolatitudes of the NCB and India were calculated from the reference poles of Van der Voo et al. (2015) and Torsvik et al. (2012), respectively. Geomagnetic polarity column is based on Ogg et al. (2016). (b–e) The corresponding paleogeography of the NQT and adjacent blocks in the Tethyan realm during this period. The reconstruction uses Gplates. The paleolatitudes of the NQT at ~297 Ma, ~259 Ma, and ~209 Ma were determined by Song et al. (2017), this study, and Song et al. (2015), respectively.

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paleolatitude of the NQT in the Late Permian is similar to that of the coeval South China block, which was located at an equatorial paleolatitude (Huang et al., 2018; Metcalfe, 2013). The similar paleolatitudes are consistent with the similarities of shared Permian fauna and flora (He et al., 2009; Zhang et al., 2013), demonstrating that they were in close proximity at that time (Metcalfe, 2013; Zhao et al., 2018). The proximity is further supported by the magmatism due to the interaction between the Emeishan plume and the Paleo-Tethyan subduction system (Liu et al., 2016; Wang et al., 2018). These pieces of evidence, together with a continental rift environment of the Late Permian mafic volcanic sequences in the NQT, seems to indicate that the NQT and SCB have been contiguous until the Late Permian when the Emeishan plume triggered their breakup (e.g., Chung et al., 1998). Because the Late Permian bimodal volcanic rocks and oceanic-island basalt-like mafic rocks related to the plume-subduction interaction were identified in the northeastern margin of the NQT (e.g., Liu et al., 2016; Wang et al., 2018), the NQT was probably located on the southwest of the Emeishan large igneous province of the SCB based on our paleomagnetic results (Figures 3a and 3d). Additionally, the paleolatitude of the NQT is similar to that of the southern margin of the SCB at the Early Permian (Huang et al., 2018; Song et al., 2017; Zhao et al., 1996). All these scenarios imply a continental connection between the NQT and the SCB as early as the Early Permian. To summarize, the paleomagnetic, paleontological, and magmatic constraints indicated that the NQT moved northward together with the SCB during the Permian.

4.4. Further Implication for the Origin of the NQT

Song et al. (2017) suggest that the NQT rifted from the Greater India margin of Gondwana prior to the Permian and moved northward during the Permian and Triassic periods based on the paleomagnetic constraints at this time. The further question is how did it rift from the Greater India margin of the Gondwana. Did it move northward together with the SQT? No reliable paleomagnetic results are available to constrain the paleolatitude of SQT at the Late Paleozoic. However, the Permo-Carboniferous strata of the SQT are characterized by glaciomarine deposits and cold-water biota, implying a typical Gondwana affinity (e.g., Zhang et al., 2013). This means that the SQT was still located at the northern margin of the Gondwana in the Early Permian. However, at the same time, the NQT drifted away from the northern margin of Gondwana (Song et al., 2017). The paleogeographic reconstruction in the Early Permian suggests that the NQT and the SQT were separated by an ancient ocean (Zhao et al., 2018; Figure 3c). The existence of this ocean has been confirmed by the Permian and Carboniferous ophiolites in central Qiangtang (Zhang et al., 2016). This conclusion is consistent with the markedly different magmatism in the NQT and the SQT during the Late Paleozoic to Triassic (Zhai et al., 2013). The Triassic arc magmatism exposed in the southern margin of the NQT was attributed to the northward subduction of the Longmu Co-Shuanghu Ocean. Notably, no arc-type magmatism can be found in the SQT during the Late Paleozoic to Triassic (Zhai et al., 2013). The magmatism of the SQT was related to a mantle plume, which may have triggered the initial rifting of the SQT from the Gondwana margin (Liao et al., 2015). These analyses indicate that the NQT was separated from the SQT when it rifted from the northern margin of Gondwana. Because the SQT was strongly influenced by the cold- to cool-temperate conditions and warm climate conditions throughout the Permian, it was regarded as a *drifting median plate*, drifting northward from Gondwana to Cathaysia during the Permian (Zhang et al., 2013). However, few reliable paleomagnetic results are available to constrain how the SQT drifted and accreted to the NQT. Notably, Song et al. (2012) reported paleomagnetic data from the NQT and the SQT with positive fold and reversal tests and obtained similar paleolatitudes (~29°N) for both in the Late Triassic, which indicates that the NQT and the SQT formed a single unit at least as early as the Late Triassic (Song et al., 2012; Figure 3e). This is consistent with the age (~233 Ma) of the eclogite-facies metamorphism of Pianshishan eclogites in central Qiangtang (Dan, Wang, White, et al., 2018).

When did the NQT rift off Gondwana? The paleolatitude difference of ~32.6° between the NQT and the expected Gondwana at the earliest Permian suggests that the NQT rifted off Gondwana prior to the Permian (Figure 3a). The paleolatitude evolution at Permian shows that the NQT moved northward at a rate of ~4.2 cm/year during ~297–259 Ma. Extrapolating a constant northward velocity has been widely used in tectonic applications of paleomagnetism (e.g., Lippert et al., 2014). The paleolatitude evolution of the Gondwanan-Indian northern margin prior to 320 Ma remains poorly constrained. Assuming that the NQT maintained a stable paleolatitude before it drifted away from the Greater India margin of Gondwana, the paleolatitude difference between the NQT and Gondwana would decrease to zero at

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~382 Ma when extrapolating the rate of ~4.2 cm/year during the Permian to pre-Permian, implying dispersion of the NQT at this time. Given the large uncertainty of the paleolatitude evolution of Gondwana, we tentatively concluded that the NQT rifted off the Greater Indian margin of Gondwana in the Devonian (Figures 3a and 3b). This conclusion is consistent with that the Late Devonian volcanic arc developed in the southern margin of the NQT (Dan, Wang, Zhang, et al., 2018). Notably, more reliable paleomagnetic data from Devonian and Carboniferous strata of the NQT and the Greater India margin of Gondwana are critical to constrain the rifting process.

4.5. Conclusions

We have provided robust and well-dated paleomagnetic results from the Late Permian Nayixiong Fm volcanics in the NQT. These data average out secular variation and thus can be used for reconstruction. Our new results show that the NQT was located at equatorial paleolatitude ($-7.6 \pm 5.6°$N) at the late Permian (~259 Ma). Combined with paleontological and magmatic constraints, our results reveal that the NQT drifted northward together with the SCB during the Permian and that the NQT drifted away from the northern margin of Gondwana in the Devonian, which is earlier than the departure time of the SQT.

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